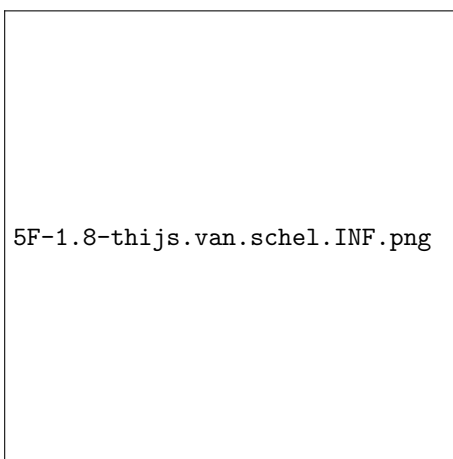


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Oefeningenreeks 5F nr. 1.8

$$r = 2(1 - \sin(\theta))$$

$$\begin{aligned} & \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta \\ &= \frac{1}{2} \int_0^{2\pi} (2 - \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_0^{2\pi} 2^2 - 8 \sin \theta + 4 \sin^2 \theta d\theta \\ &= 2 \int_0^{2\pi} d\theta - 4 \int_0^{2\pi} \sin \theta d\theta + 2 \int_0^{2\pi} \sin^2 \theta d\theta \\ &= 2(2\pi) - 4(0) + 2 \int_0^{2\pi} \frac{1 - \cos(2\theta)}{2} d\theta \\ &= 4\pi + 2 \cdot \frac{1}{2} \int_0^{2\pi} d\theta - 2 \cdot \frac{1}{2} \int_0^{2\pi} \cos(2\theta) d\theta \\ &= 4\pi + 2 \left(\frac{1}{2} (2\pi - 0) \right) \\ &= 4\pi + 2\pi = 6\pi \end{aligned}$$

Tekening:



References